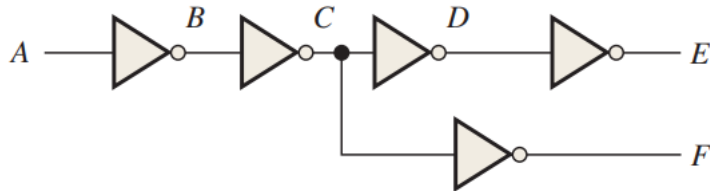


Chapter 3

2024.2.22

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2. A combination of inverters is shown in Figure 3-60. If a HIGH is applied to point A, determine the logic levels at points B through F.



Answer Area:

Because the inverters are all NOT gate. Therefore

B: LOW

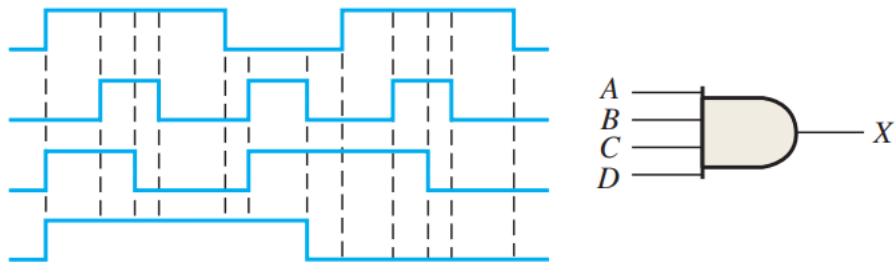
C: HIGH

D: LOW

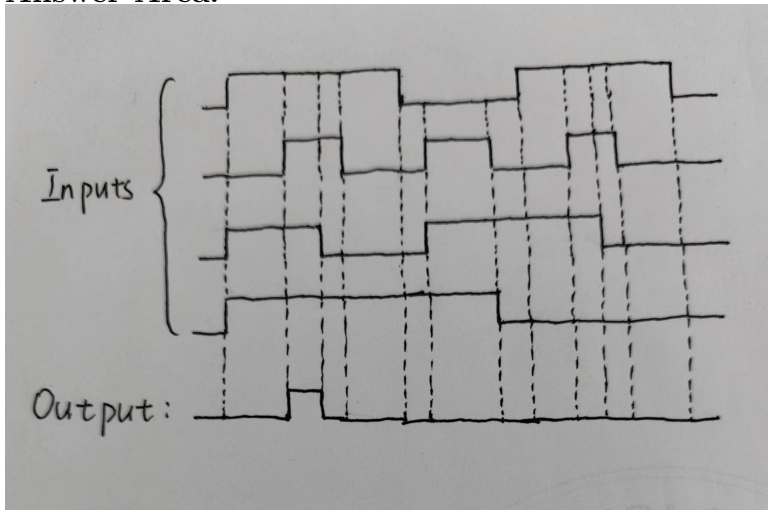
E: HIGH

F: LOW

8. The input waveforms applied to a 4-input AND gate are as indicated in Figure 3-64. Show the output waveform in proper relation to the inputs with a timing diagram.

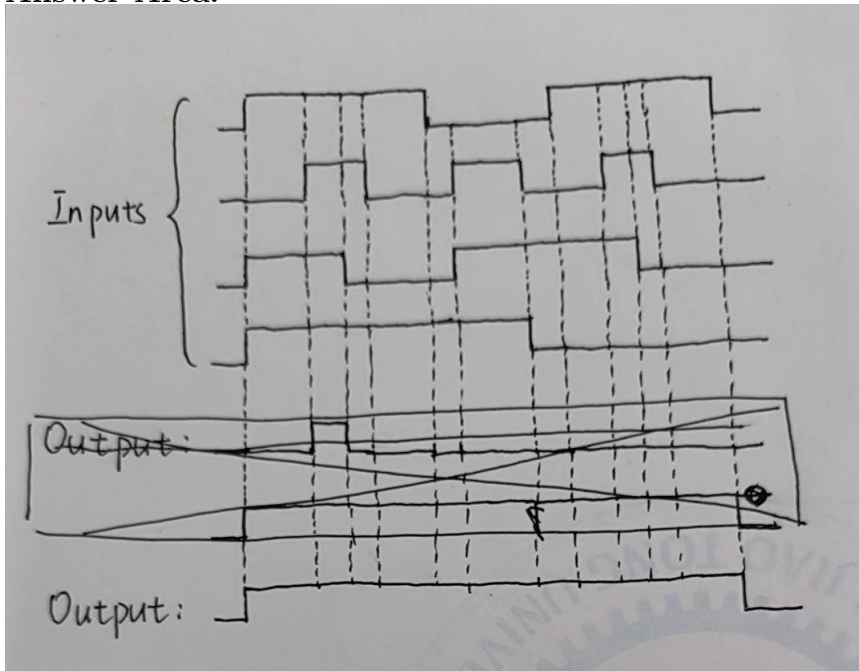


Answer Area:



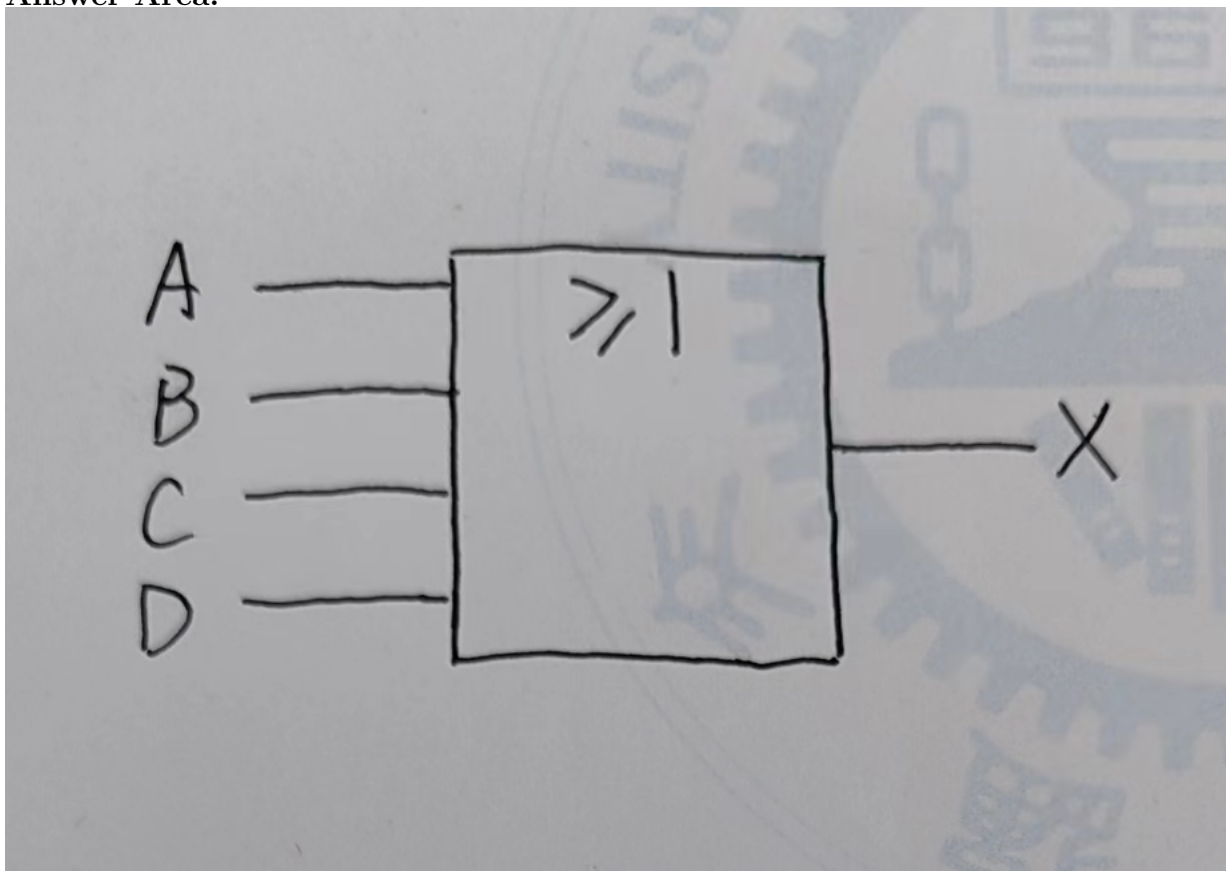
13.Repeat Problem 8 for a 4-input OR gate.

Answer Area:

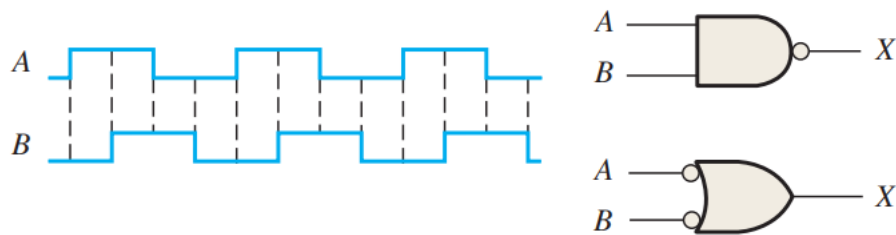


15.Draw the rectangular outline symbol for a 4-input OR gate.

Answer Area:

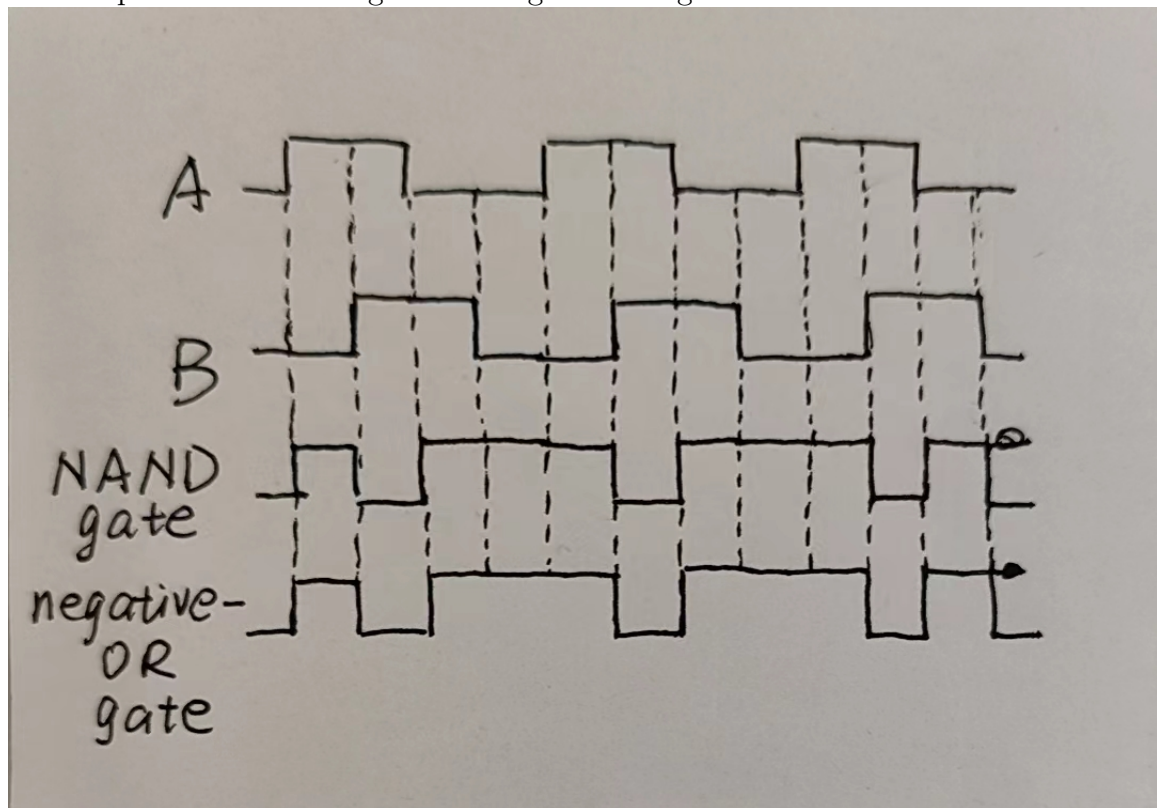


20. As you have learned, the two logic symbols shown in Figure 3-69 represent equivalent operations. The difference between the two is strictly from a functional viewpoint. For the NAND symbol, look for two HIGHS on the inputs to give a LOW output. For the negative-OR, look for at least one LOW on the inputs to give a HIGH on the output. Using these two functional viewpoints, show that each gate will produce the same output for the given inputs.



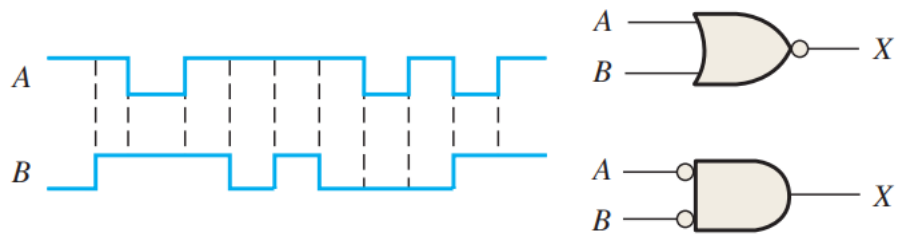
Answer area:

The output of the NAND gate and negative-OR gate are as follows:



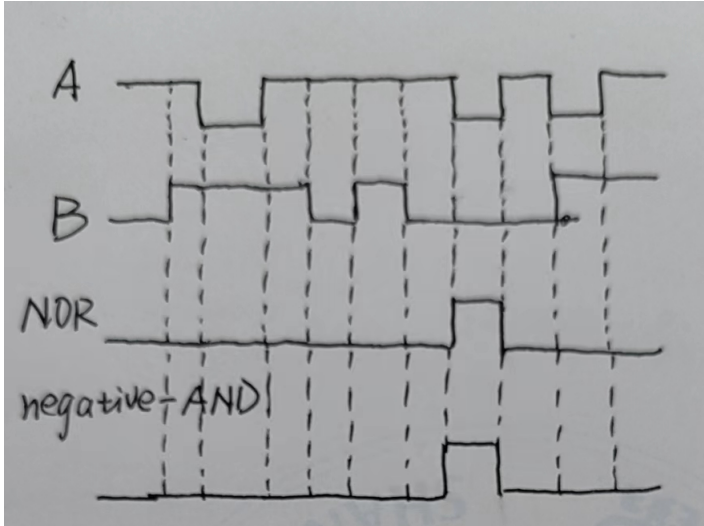
And the two outputs are the same.

24. The NAND and the negative-OR symbols represent equivalent operations, but they are functionally different. For the NOR symbol, look for at least one HIGH on the inputs to give a LOW on the output. For the negative-AND, look for two LOWs on the inputs to give a HIGH output. Using these two functional points of view, show that both gates in Figure 3-71 will produce the same output for the given inputs



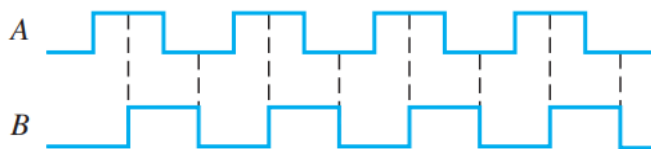
Answer Area:

The output of the NOR gate and negative-AND gate are as follows:



And the two outputs are the same.

28. Determine the output of an exclusive-OR gate for the inputs shown in Figure 3-62 and draw a timing diagram.



Answer Area:

